

Assignment 01: Applied AI

Word Ladder Adventure – A Game-Based Implementation Using Search Algorithms

Project Overview

This project challenges you to develop a **Word Ladder Adventure Game**, where players must transform one word into another by changing a single letter at a time. Each intermediate step in the transformation must be a valid word from a predefined dictionary. To enhance the gameplay experience, the game will incorporate at least **three different search algorithms** to compute the optimal transformation path. Players will have the choice to either solve the puzzles manually or seek assistance from an **AI-powered hint system**, which suggests the shortest transformation path based on the chosen search algorithm. Through this project, you will gain hands-on experience with **graph-based search algorithms, heuristic functions, and AI-driven decision-making**, while also building an engaging and interactive puzzle-solving application.

Game Concept & Features

The **Word Ladder Adventure Game** revolves around transforming an initial word into a target word by making incremental one-letter changes, where each intermediate word must be valid according to the dictionary. The player can either choose a predefined set of word ladder challenges or input a custom starting and ending word, provided that a valid transformation path exists between them. The goal is to complete the transformation in as few steps as possible, with the number of transformations determining the player's final score. The fewer the moves taken, the higher the score.

The gameplay experience is designed to be both **manual and AI-assisted**. Players can attempt to solve the word transformation themselves or request hints from an **AI assistant** that employs different search algorithms to suggest the next best move. The game interface will feature a **graph-based visualization**, where words are represented as nodes and valid transformations are displayed as edges connecting them. Additionally, the game imposes a limit on the number of moves a player can take, increasing the challenge and strategic depth.

AI Assistance & Search Algorithms

The game will integrate an **AI-powered hint system** that finds the optimal path between the start and goal words by treating words as nodes in a graph and valid transformations as edges. The AI assistant will be equipped with at least **three search algorithms**, allowing players to compare their performance in finding the shortest word transformation sequence. The three algorithms that can be implemented are *Uniform Cost Search (UCS)*, *A Search Algorithm*, and either Breadth-First Search (BFS) or Greedy Best-First Search (GBFS).

Players will have the option to select an algorithm when requesting hints, making the game an **educational tool for understanding search strategies** in artificial intelligence.

Each algorithm will follow a structured search strategy, with the core functions guiding the AI's decision-making process.

1. The **$g(n)$ function** will define a valid cost function, determining how much effort has been spent to reach the current word.
2. The **$h(n)$ function** will serve as a heuristic, estimating how many more transformations are needed to reach the target word.
3. Finally, the **$f(n)$ function**, defined as **$f(n) = g(n) + h(n)$** , will be used to prioritize the best transformation path, ensuring an optimal solution is found efficiently.

Game Modes & Difficulty Levels

To provide a well-rounded experience for different types of players, the game will feature multiple difficulty levels and game modes.

- 1. In **Beginner Mode**, players will encounter simple word ladders with short transformation sequences, such as changing "cat" to "big".
- 2. The **Advanced Mode** will introduce longer and more complex word transformations, such as converting "stone" to "money", requiring deeper strategic thinking.
- 3. Lastly, a **Challenge Mode** will introduce obstacles such as **banned words, restricted letters, or additional constraints**, making the puzzle-solving process more dynamic and engaging.

Technical Requirements

- **Programming Language:** Python
- **Data Structure:** Graph-based representation of words
- **Algorithm:** 3 different search algorithms for finding the shortest path
- **Dictionary Source:** Can be a preloaded text file or an external API

Project Deliverables

- 1. **Game Implementation:**
 - Functional **Word Ladder Adventure Game**.
 - A working Search-based AI assistant for hint generation.
- 2. **Report:**
 - A report detailing the approach, algorithm, and challenges faced during development.
- 3. **Bonus Features (Optional):**
 - Multiplayer mode with live competition.

Custom word ladder creation by users

Grading Criteria

Component	Marks
Game Functionality	40%
Search Algorithm Implementation and comparisons	30%
UI/UX & Game Design	15%
Documentation & Report	10%
Bonus Features	5%